Artificial Intelligence in Team Sports: Methods for Data Collection and Performance Analysis – A Narrative Review

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***Abstract*—AI is now central in sports analytics, helping to improve player monitoring, identification and assessment of their performance. This paper looks closely at AI strategies for football, with major emphasis on machine learning, neural networks and computer vision approaches. By reviewing articles, we study how AI contributes to ongoing player monitoring, prediction of events and decision-making during matches. The results indicate progress in automated data handling, but certain problems persist, including the need for understandable algorithms, the question of who owns the data, biased results and adjustment to changing situations. Working on these limitations will give sports the best chance to use AI. Future research should emphasize the development of adaptive AI models, the integration of physiological and psychological factors, and ethical considerations related to data security and transparency. By overcoming these obstacles, AI can drive a new era in sports analytics, providing deeper insights and optimizing team performance in an increasingly data-driven competitive landscape.**

Keywords—AI player tracking, Real-time identification, Sports Data analysis, Computer vision football, Deep learning, Video analysis AI, Machine learning tactics, Performance prediction, Ethical AI in Sports.

I. Introduction

Recently, artificial intelligence has revolutionized many sectors and sports is one of them [1]. Because it can quickly and accurately handle large amounts of data, AI creates new chances to improve athletes, craft improved strategies and delight sports fans. With the help of AI, real-time player tracking, on-game action analysis, predicting match outcomes and monitoring athletes' health have been developed for team sports [2].

Technologies such as visual recognition, neural networks and machine learning, it has become much easier to observe player activities, expect what will happen on the field and support coaches and referees when making decisions. Even though AI provides many important advantages to sports, there are still some difficulties to overcome. A major barrier is that both AI models and their data can be very complex [3]. As an example, player tracking in real time needs advanced computer vision tools able to get around issues like obstructions and variations in light. Other concerns are related to data secrecy, managing biases in AI and how teams can ensure that AI-based decisions are clear. Because of these challenges, we must work carefully to use AI for team sports, so that it improves play and reduces the risks involved.

The study seeks to explore which AI techniques are being used today to gather and review statistics in team sports. By means of a narrative review, it goes over current strategies, what is advantageous about them, what holds them back and the obstacles to their ethical use. We set out to assess the current AI methods in team sports by highlighting the most useful approaches for both instant location monitoring of players and predicting their performance, comparing the main features of CNNs, YOLO, RNNs and SVM models and pointing out gaps in research that could be utilized for future work.

The rest of this paper is organized as follows: the second section outlines the methodology for selecting and reviewing studies, the third section presents the results on AI techniques in football performance analysis, player tracking, and sport data match analysis, the fourth section discusses challenges, limitations, and ethical considerations, and the fifth section concludes with a summary of findings and suggestions for future research.

II. Methods

This section details the methodology used to identify, select, and assess studies on the application of AI in sports. It describes the search strategy, inclusion criteria, quality assessment, and thematic classification of the selected studies.

A. Procedures and Search Strategy

A narrative literature review was conducted to examine the application of artificial intelligence in team sports performance analysis. The search was carried out exclusively in the Scopus

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database, selected for its broad coverage of peer-reviewed and multidisciplinary research. The study selection process was guided by predefined eligibility criteria focusing on the relevance, quality, and methodological rigor of each article. Publications addressing AI-based methods for player tracking, identification, and performance prediction in football were prioritized. The detailed inclusion and exclusion process is illustrated in Figure 1.

B. Eligibility Criteria and Selection Process

Studies were included based on predefined criteria related to language, publication type, timeframe, and thematic relevance to AI applications in sports performance. A three-stage selection process—screening, full-text review, and independent evaluation—was conducted to ensure the quality and consistency of the final dataset.

The overall study selection procedure is illustrated in the PRISMA flow diagram presented in Figure 1.

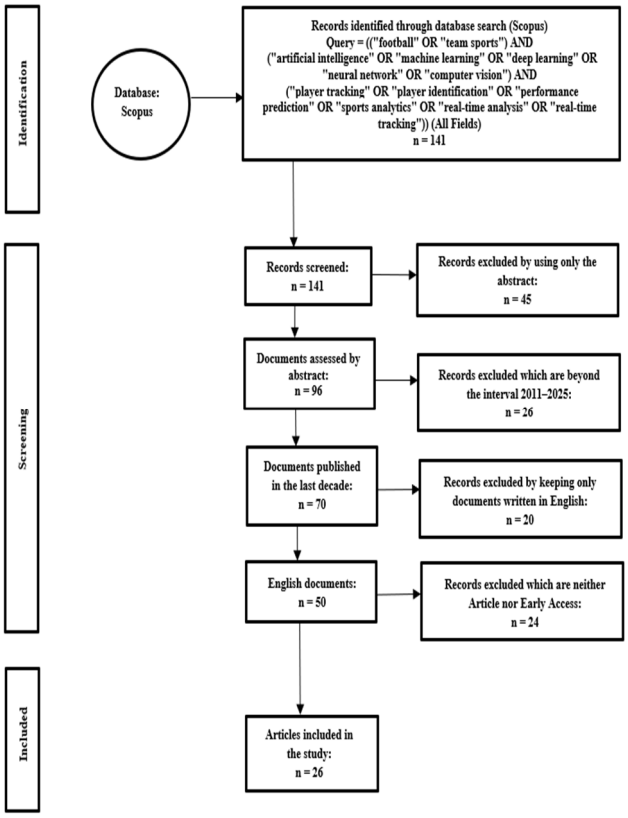

Figure 1. PRISMA flow diagram illustrating the article selection process

C. Quality Assessment and Theme Classification

To ensure methodological rigor, the quality of the selected studies was evaluated using a 7-point scoring system based on the following criteria:

- Methodological validity (robustness of the study design and AI techniques used).
- Data transparency (clarity and availability of datasets).
- Reproducibility (ability to replicate the findings).
- Applicability (relevance of the conclusions for sports performance enhancement).

Studies scoring $\geqslant 5$ were considered to have low risk of bias, whereas those scoring $\leqslant 4$ were excluded from the final analysis [5]. Additionally, backward and forward citation tracking was performed to identify any further relevant studies [6].

Finally, the selected studies were categorized into key thematic areas, including: AI-driven real-time player recognition, Real-time player tracking using artificial intelligence, AI-based methodologies for sports data analysis.

III. Results

In this section, we extend on the selected articles' main findings to subthemes to get insights into how AI is adopted in sports. Each sub-theme addresses a specific aspect of the research, with relevant articles cited to support the discussion.

A. AI-driven Real-time Player Recognition

This study [7] enhances YOLOv5 for real-time football analysis by integrating SimSPPF, a Simplified Spatial Pyramid Pooling Fast module that enables efficient multi-scale feature extraction; GhostNet, a lightweight neural network module designed to reduce computational complexity by generating ghost feature maps; and Slim Scale Detection Layer, an optimized layer for precise bounding box predictions, which reduces computational requirements by omitting large-scale feature maps. The research made in [8] proposes a method combining spatial constellation-based recognition, which models the relative positions of players within their team to capture tactical patterns, and convolutional neural networks (CNNs), trained to recognize jersey numbers directly from pixel data, for jersey number recognition. This combination improves accuracy from 0.69 to 0.82 by optimally matching player trajectories with identities through the fusion of these techniques. In [9], we present method uses a Linear Programming (LP) Relaxation algorithm combined with Conditional Random Field (CRF) and weakly supervised learning to improve player identification accuracy in sports videos. It achieves similar results with only 200 labeled examples, reducing the need for large datasets. The authors of [10] propose method uses deep convolutional neural networks (CNNs) and multi-scale pooling, along with Convolutional Pose Machines (CPMs), to identify players by extracting global and local features. It achieves 96% accuracy, even under challenging conditions, through Fisher vector encoding and a SVM classifier. As per the study in [11], the Individual Level Sports Video Indexing (ILSVI) method recognizes players by their jersey numbers using a principal axis-based contour descriptor, color model, Histogram of Oriented Gradients (HOG), a feature extraction technique that captures local shape and edge information and visual saliency to highlight important regions. It achieves an 83.74% recognition rate, effectively identifying players under varying conditions.

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B. Real-time Player Tracking Using Artificial Intelligence

The study presented in [12] proposes the FOOTBALLTrace method for player tracking uses YOLOv7 for detection and a neural network for ID association, ensuring continuous player trajectories even during occlusions. It adjusts interrupted trajectories through geometric data corrections for accurate tracking. The research presented in [13] proposes a method that uses a computer vision-based approach with an occupancy map and spatio-temporal bilinear model to track players without sensors, addressing occlusion and lighting challenges. Validation was performed on a hockey field with 8 HD cameras and 200,000 images, compared against manual annotations. The authors of [14] propose The SmartCrop method leverages AI for video cropping optimization in sports, focusing on players and the ball. It adapts to various aspect ratios, improving tracking and user experience, with proven effectiveness in media presentation. In [15], we present the Markov Chain Monte Carlo (MCMC) method for data association to track players in football videos, employing unsupervised learning and a boosting classifier for detection. It ensures accurate tracking despite occlusions, camera motion, or pose variations. The authors of [16] propose a particle filter method for player tracking, where each player is represented by particles modeling possible positions. The particle positions are recalculated based on visual features, with higher-weight particles predicting future player positions. The research presented in [17] reveals a method for multi-object tracking (MOT) in football videos combines DeepSORT (Deep Simple Online and Realtime Tracking), FairMOT, and ByteTrack algorithms to track players, the ball, and referees, addressing challenges like occlusions and rapid movements. It introduces the SoccerNet-Tracking dataset, the largest for MOT in football, to test these methods in real match scenarios.

C. Methodology for sports data analysis using AI

This study [18] uses machine learning to extract insights from complex sports data, focusing on managing dependencies and contextual factors. It combines traditional sports expertise with modern data analysis in a structured framework to enhance decision-making. The researchers in [19] propose a method named a deterministic decision tree algorithm to automatically extract football events from tracking data by detecting ball possession and player configurations. It achieved over 90% detection accuracy across various tournaments and data providers. In [20], the authors propose the ISPMA (Intelligent Strategic Planning Method-Based Algorithm) to analyze match statistics and predict outcomes. It selects the most relevant data (shots on target, halftime goals, and total shots) using advanced feature selection techniques (Extra Tree Classifier, Forward Selection, Chi-square Test, Pearson Correlation) and processes them with machine learning models (Random Forest, Support Vector Machine (SVM), Naïve Bayes). The algorithm predicts match outcomes with 98.80% accuracy and provides real-time strategic recommendations to optimize team performance. In [21], we present a method that uses positional data and Big Data technologies to develop advanced KPIs for soccer, focusing on real-time analysis of player and team performance. It aims to predict match outcomes and enhance training through dynamic tracking and tactical insights. The study in [22] applies Multitask Ridge Regression (MT-RR) and Multitask Support Vector Regression (MT-SVR) to predict football player performance based on historical match data. It evaluates multiple players simultaneously, improving prediction accuracy by considering individual characteristics. The research made in [23] uses Support Vector Machines (SVM) and Artificial Neural Networks (ANN) to automatically analyze and summarize soccer videos. It segments the video into shots and classifies them by type. Important events, such as goals, are detected through logo and score detection, as well as by analyzing goalposts and the net using k-means clustering and Hough transform. The authors of [24] present a fully convolutional neural network (FCN) architecture for analyzing soccer. This model estimates pass probability surfaces using high-frequency spatiotemporal data. It learns features at different spatial scales, enabling the prediction of pass success probabilities and the analysis of player positioning and decision-making. The research made in [25] employs deep learning methods, including CNNs like ResNet and temporal aggregation techniques like NetVLAD, to analyze broadcast soccer videos. These methods enhance action detection, sequence segmentation, and replay identification by leveraging contextual temporal information. This study [26] employs Random Forest, Decision Tree, and Naïve Bayes algorithms to predict football players' physical performance.

In conclusion, the research demonstrates improvements in player identification, activity monitoring and analyzing sports data, thanks to AI despite challenges posed by objects in front of the cameras and data complexity. As a result, players can be tracked live, providing useful information about their gameplay and matches. The next part will examine the larger significance of these findings.

IV. Discussion

This discussion we will look at how identification and tracking of players is being carried out with the latest video analysis in sports. It will discuss the approaches scientists use, main results achieved and obstacles they encountered applying them, pointing out their ability to increase the quality of sports performance analysis and better understand match situations.

Table 1. Summary of AI and Computer Vision Techniques Applied to Sports Video Analysis

Reference	Technique	Methodology	Findings	Challenges
[7]	Player and ball detection	YOLOv5 with SimSPPF, GhostNet, and Slim Scale for improved feature extraction and real-time detection	Improved player and ball detection performance in complex scenarios	Requires high-quality data; challenges with occluded players and crowded matches
[8]	Player identification via	Combines player position	Improved player identificat	Complexity in optimizing

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	position and jersey number	recognition (spatial constellation) and jersey number recognition via CNNs for player identification in soccer videos	ion accuracy from 0.69 to 0.82 by combining both techniques	trajectory assignment to player identities and managing temporal and spatial data
[9]	Player Identification	Combine weak visual indices, temporal constraints, mutual exclusion via CRF, and use play-by-play texts to reduce label requirements	Similar accuracy with only 200 labels, compared to 20,000 for strong learning	Managing occlusions and complex player movements
[10]	Player Identification	Uses multi-scale CNN features and Fisher Vector encoding to identify players in broadcast videos	Achieves 96% accuracy in NBA basketball clips, able to identify players in difficult scenes	Ambiguity in player appearance, body parts covered by jerseys, and challenges with localizing distinguishing features
[11]	Player recognition via jersey numbers	Use of principal axis-based contour descriptors, HOG, color model, and visual saliency to extract numbers.	Recognition rate of 83.74%, effective under varying conditions (scales and rotations)	Sensitive to blurry videos, low resolution, and poor image quality
[12]	Player Tracking	FOOTBALLTrace uses YOLOv7 for detection and a neural network for ID association, focusing on trajectory continuity and occlusion handling	Maintains player trajectory even during partial occlusions	Handling occlusions and interrupted trajectories
[13]	Player Tracking	Computer vision-based approach using occupancy map and spatio-temporal bilinear model to track players without sensors	Achieved accurate player tracking, validated with manual annotations	Challenges with occlusion and varying lighting conditions
[14]	Player Tracking	SmartCrop method for video cropping optimization, focusing on key elements like players and bal	Adapts to various aspect ratios, enhances media presentation	Handling various aspect ratios and scene transitions
[15]	Player Tracking	Uses Markov Chain Monte Carlo (MCMC)for data association across frames with unsupervised learning for player detection.	Accurate player tracking even with occlusions and camera motion.	Handling occlusions and false detections.
[16]	Player Tracking	Uses Particle Filters for tracking, Deformable Part Model (DPM) for player detection, and OCR for jersey number recognition	82% precision in tracking, 79% precision, and 77% recall in identification.	Occlusions, blurred faces, ambiguous appearances, and motion complexities in dynamic settings.
[17]	Player Tracking	Use of DeepSORT algorithm with Kalman Filtering and the Hungarian algorithm for association	DeepSORT approach provides good performance for object association, but requires ground truth detections for better results	Challenges with fast movements and occlusions, especially in complex football match situations.
[18]	Sports Analytics, Model Evaluation, Data Integration	Machine Learning (ML) used to extract actionable insights from complex data. Focus on managing dependencies and contextual factors.	Actionable insights for sports performance. Combining traditional sports expertise with data analysis.	Managing dependencies and contextual factors. Difficulty separating indicators from models.
[19]	Match statistics analysis	CNN for video analysis, RNN for time-sequenced events, NLP for match report analysis.	High accuracy in detecting match events (goals, passes, shots), effective text analysis for match insights.	Requires large datasets, challenges in video synchronization, variability in play strategies.

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[20]	Football match statistics analysis	Uses ISPMA (Intelligent Strategic Planning Method-Based Algorithm), which applies feature selection techniques (Extra Tree Classifier, Forward Selection, Chi-square, Pearson Correlation) and Machine Learning models (Random Forest, SVM, Naïve Bayes, etc.) to predict match outcomes based on first-half statistics..	Achieves 98.80% accuracy, generates real-time strategic recommendations, and predicts match outcomes at halftime.	Data limited to the English Premier League, requires validation on other leagues, and real-time adaptation is challenging.
[21]	Modern Match Analysis in Soccer	Use of positional tracking and Big Data with neural networks for soccer performance analysis.	Advanced KPIs based on positional data provide deeper insights into team and player performance. Big Data allows for real-time and dynamic analyses.	Insufficient validation of new KPIs, lack of infrastructure for Big Data, bridging theory and practice.
[22]	Analysis of Soccer Player Performance	Multitask regression (ridge regression and support vector regression) to predict player performance. Feature selection using recursive feature elimination (RFE).	Multitask models performed better than baselines. Best performance achieved with 8-game windows.	Data sparsity and cold-start problem for new players.
[23]	SoccerVideo Analysis, Classification, Event Detection	Segmenting the video into small shots. - Classifying video shots (long, medium, close-up, audience). - Replay detection using SVM and ANN. - Detecting exciting events (score, goal posts, net) using K-means, Hough transform. - Summarizing major events.	Good performance in precision and recall, with SVM preferred.	Video format variability and classification difficulty.
[24]	Visually interpretable analysis in soccer	Fully Convolutional Neural Network (FCN) for learning pass probability surfaces using spatiotemporal soccer data.	Provides interpretable insights for pass success, optimal positioning, and passing tendencies.	Addressing weak supervision, spatial feature integration, and scalability to broader contexts.
[25]	Benchmarking for Soccer Video Analysis	Deep Learning (ResNet, NetVLAD, CALF), CNNs for feature extraction and temporal aggregation	Improved action detection, sequence segmentation, and replay identification	Data imbalance, complexity in detecting off-screen actions
[26]	Predicting players' physical performance during a football match	Random Forest, Decision Tree, Naïve Bayes	Random Forest provides the best accuracy, while Decision Tree and Naïve Bayes are simpler but offer lower accuracy.	Managing imbalanced data (underperforming vs. performing classes) and the complexity of variable data

A. Taxonomy of AI Applications in Team Sports

To better clarify AI uses and meet the needs for integration pointed out in recent critiques, we present a taxonomy of AI applications focusing on team sports analytics. Its structure is defined by three main working domains:

- **Player Detection and Recognition** – encompassing methods such as YOLOv5 and convolutional neural networks (CNNs), which identify players based on visual features, jersey numbers, or spatial positioning.
- **Real-Time Tracking** – involving approaches like DeepSORT, Particle Filters, and Markov Chain Monte Carlo (MCMC), which ensure continuous tracking of players across frames despite occlusions and movement variability.
- **Performance Prediction and Tactical Analysis** – applying models such as Support Vector Machines (SVMs), Random Forests, Fully Convolutional

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Networks (FCNs), and Transformer-based architectures to forecast outcomes, analyze team behavior, or suggest strategy optimization.

This taxonomy not only highlights the key technical pathways currently employed but also serves as a practical guide for selecting appropriate methods depending on specific analytical objectives within sports environments.

We summarize the taxonomy in Table 2 below:

Table 2. Functional Taxonomy of AI Techniques in Team Sports Analytics

Category	AI Techniques	Strengths	Limits
Detection & Recognition	YOLOv5, CNNs	High accuracy, efficient real-time detection	Sensitive to low video quality and occlusions
Real-Time Tracking	DeepSORT, MCMC, Particle Filters	Robust under occlusion, maintains trajectories	Requires high-quality detection; complex to train
Prediction & Tactical Analysis	SVMs, FCNs, Transformers, Random Forest	High interpretability, enables strategy optimization	May require large datasets; less real-time capable

The analysis of the results highlights the growing effectiveness of artificial intelligence methods for player detection, identification, and tracking in team sports. Deep learning models like YOLOv5 and DeepSORT consistently outperform traditional models, especially under tough conditions where players are packed close together or there are some occlusions. These methods, however, struggle to generalize well across different match conditions, and require largely to high quality and annotated data. On the other hand, traditional models such as SVMs and Decision Trees are still common in specific formulations—like injury prediction or performance classification—given their interpretability and their less complex computational requirements (even when they may not achieve the same modeling quality of deep models), though they typically do not possess the flexibility and expressiveness of deep models. However, current systems still have a number of limitations in terms of real-time adaptation, cross-league convergence, and robustness under low quality video or high dynamics conditions. The presented approaches indicate slightly higher degree of integration and accuracy with respect to previously published methods, nevertheless they still require further validation. These AI developments are directly applicable to player monitoring, tactical optimization and AI-assisted refereeing, however future work is required to create adaptive, hybrid systems and to promote ethical and effective integration of these technologies during live matches.

However, a number of voids still exist despite these technological improvements. First, not enough physiological and psychological measures are integrated into AI systems to support whole player analysis. Second, there is scarce literature that considers the ability of the AI model to adapt its predictions when changing leagues, game intensities, or weather conditions. Third, a range of the ethical and legal elements, including ownership of data, consent, and transparency, remain under- to unexplored in practice. These lacunae are also important directions for future research.

V. Conclusion

This paper underlines a breakthrough design and development in Football Performance Analysis using Artificial Intelligence based on advanced Computer Vision, Deep Learning and Predictive Analytics approach to enrich the player tracking, recognition and decision making. But several obstacles, including lack of interpretability and data security and the need for real-time agility, need to be overcome in order to fully harness the power of AI. Future research directions to be addressed include hybrid AI models with integration of physiological and psychological data, robustness enhancement of AI systems in complicated match environments, ethical AI adoptions by building regulatory guidelines, understanding the role of AI in injury prevention by setting biophysical and biomarker analysis, real-time AI decisions in dynamic sports scenarios, among others. Solving for these, AI stands to be revelatory in sports analytics, giving teams, players, and analysts actionable feedback.

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References

[1] P. Ding, “Analysis of Artificial Intelligence (AI) Application in Sports,” presented at the Journal of Physics: Conference Series, 2019. doi: 10.1088/1742-6596/1302/3/032044.

[2] F. J. J. Reis, R. K. Alaiti, C. S. Vallio, and L. Hespanhol, “Artificial intelligence and Machine Learning approaches in sports: Concepts, applications, challenges, and future perspectives,” Brazilian Journal of Physical Therapy, vol. 28, no. 3, 2024, doi: 10.1016/j.bjpt.2024.101083.

[3] S. Afzal et al., “Visualization and Visual Analytics Approaches for Image and Video Datasets: A Survey,” ACM Transactions on Interactive Intelligent Systems, vol. 13, no. 1, 2023, doi: 10.1145/3576935.

[4] U. Aiswarya and S. Binthu Mathavan, “Artificial Intelligence in Sports Performance Evaluation of Team Games,” presented at the ACM International Conference Proceeding Series, 2023. doi: 10.1145/3647444.3647914.

[5] A. E. Saw, L. C. Main, and P. B. Gastin, “Monitoring the athlete training response: Subjective self-reported measures trump commonly used objective measures: A systematic review,” British Journal of Sports Medicine, vol. 50, no. 5, pp. 281–291, 2016, doi: 10.1136/bjsports-2015-094758.

[6] T. Horsley, O. Dingwall, and M. Sampson, “Checking reference lists to find additional studies for systematic reviews,” Cochrane Database of Systematic Reviews, vol. 2011, no. 8, 2011, doi: 10.1002/14651858.MR000026.pub2.

[7] K. Zhao, “Enhancing the Performance and Accuracy in Real-Time Football and Player Detection Using Upgraded YOLOv5 Architecture,” International Journal of Computational Intelligence Systems, vol. 17, no. 1, 2024, doi: 10.1007/s44196-024-00565-x.

[8] S. Gerke, A. Linnemann, and K. Müller, “Soccer player recognition using spatial constellation features and jersey number recognition,” Computer Vision and Image Understanding, vol. 159, pp. 105–115, 2017, doi: 10.1016/j.cviu.2017.04.010.

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[9] W.-L. Lu, J.-A. Ting, J. J. Little, and K. P. Murphy, "Learning to track and identify players from broadcast sports videos," IEEE Transactions on Pattern Analysis and Machine Intelligence, vol. 35, no. 7, pp. 1704–1716, 2013, doi: 10.1109/TPAMI.2012.242.

[10] A. Senocak, T.-H. Oh, J. Kim, and I. S. Kweon, "Part-based player identification using deep convolutional representation and multi-scale pooling," presented at the IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, 2018, pp. 1813–1820. doi: 10.1109/CVPRW.2018.00225.

[11] S.-W. Sun et al., "Who's who in a sports video? An individual level sports video indexing system," presented at the Proceedings - IEEE International Conference on Multimedia and Expo, 2012, pp. 937–942. doi: 10.1109/ICME.2012.59.

[12] A. H. Mostafa, M. A. Rushdi, T. A. Basha, and K. Sayed, "FOOTBALLTrace: An AI-Based System for Football Player Tracking with Occlusion Detection and Trajectory Correction," presented at the International Conference on Sport Sciences Research and Technology Support, icSPORTS - Proceedings, 2023, pp. 194–201. doi: 10.5220/0012226800003587.

[13] A. Bialkowski, P. Lucey, P. Carr, S. Sridharan, and I. Matthews, "Representing team behaviours from noisy data using player role," Advances in Computer Vision and Pattern Recognition, vol. 71, pp. 247–269, 2014, doi: 10.1007/978-3-319-09396-3_12.

[14] S. M. M. Dorcheh et al., "AI-Based Cropping of Sport Videos Using SmartCrop," International Journal of Semantic Computing, vol. 18, no. 4, pp. 637–662, 2024, doi: 10.1142/S1793351X24450028.

[15] J. Liu et al., "Automatic player detection, labeling and tracking in broadcast soccer video," presented at the BMVC 2007 - Proceedings of the British Machine Vision Conference 2007, 2007. doi: 10.5244/C.21.3.

[16] C.-W. Lu, C.-Y. Lin, C.-Y. Hsu, M.-F. Weng, L.-W. Kang, and H.-Y. M. Liao, "Identification and tracking of players in sport videos," presented at the ACM International Conference Proceeding Series, 2013, pp. 113–116. doi: 10.1145/2499788.2499842.

[17] A. Cioppa et al., "SoccerNet-Tracking: Multiple Object Tracking Dataset and Benchmark in Soccer Videos," presented at the IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, 2022, pp. 3490–3501. doi: 10.1109/CVPRW56347.2022.00393.

[18] J. Davis et al., "Methodology and evaluation in sports analytics: challenges, approaches, and lessons learned," Machine Learning, vol. 113, no. 9, pp. 6977–7010, 2024, doi: 10.1007/s10994-024-06585-0.

[19] F. Vidal-Codina, N. Evans, B. El Fakir, and J. Billingham, "Automatic event detection in football using tracking data," Sports Engineering, vol. 25, no. 1, 2022, doi: 10.1007/s12283-022-00381-6.

[20] K. Madan, K. Taneja, and H. Taneja, "Intelligent Strategic Planning Method based Algorithm (ISPMA) for Estimation of Soccer Sports Match Outcome," International Journal of Mathematical, Engineering and Management Sciences, vol. 10, no. 1, pp. 249–264, 2025, doi: 10.33889/IJMEMS.2025.10.1.014.

[21] D. Memmert and R. Rein, "Match analysis, big data and tactics: Current trends in elite soccer," Deutsche Zeitschrift fur Sportmedizin, vol. 69, no. 3, pp. 65–72, 2018, doi: 10.5960/dzsm.2018.322.

[22] C. Arndt and U. Brefeld, "Predicting the future performance of soccer players," Statistical Analysis and Data Mining, vol. 9, no. 5, pp. 373–382, 2016, doi: 10.1002/sam.11321.

[23] H. M. Zawbaa, N. El-Bendary, A. E. Hassanien, and T. Kim, "Event detection based approach for soccer video summarization using machine learning," International Journal of Multimedia and Ubiquitous Engineering, vol. 7, no. 2, 2012.

[24] J. Fernández and L. Bornn, "SoccerMap: A Deep Learning Architecture for Visually-Interpretable Analysis in Soccer," Lecture Notes in Computer Science (including subseries Lecture Notes in Artificial Intelligence and Lecture Notes in Bioinformatics), vol. 12461 LNAI, pp. 491–506, 2021, doi: 10.1007/978-3-030-67670-4_30.

[25] A. Deliege et al., "SoccerNet-v2: A dataset and benchmarks for holistic understanding of broadcast soccer videos," presented at the IEEE Computer Society Conference on Computer Vision and Pattern Recognition Workshops, 2021, pp. 4503–4514. doi: 10.1109/CVPRW53098.2021.00508.

[26] T. B. Dijkhuis, M. Kempe, and K. A. P. M. Lemmink, "Early prediction of physical performance in elite soccer matches—A machine learning approach to support substitutions," Entropy, vol. 23, no. 8, 2021, doi: 10.3390/e23080952.

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